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PALEOMAGNETIC RESULTS FROM THE UPPER CRETACEOUS MAUDLOW AND LIVINGSTON FORMATIONS, SOUTHWEST MONTANA

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Abstract. A paleomagnetic investigation of welded tuff units from the Late Cretaceous Livingston and Maudlow Formations was undertaken to refine the Late Cretaceous to Paleocene APWP for cratonic North America. After progressive AF and thermal demagnetization, the Maudlow Formation (46.1°N, 248.9°E) yields a paleomagnetic pole at 70°N, 208°E (11 sites, A_{95}=9°, K=22). Tectonic correction improves the directional grouping, and the change is statistically significant. Two normal polarity intervals bounding a single reversed polarity interval are present. We interpret the reversed interval to be Chron 33r, on the basis of: [1] stratigraphic correlation of the Maudlow Formation with the Campanian Sedan Formation located 40 km to the east, and [2] K-Ar dates of 83 ± 2 Ma at the middle of the sampled section and 74.9 ± 1 Ma near the top of the sampled section.

The Livingston Formation (45.5°N, 250.1°E) yields a high H_c magnetization with a negative fold test, suggesting a post-folding secondary magnetization. The Livingston mean direction yields a paleomagnetic pole at 72° N, 218° E (10 sites, A_{95}=4°, K=107) and is clearly distinguishable from a co-existing lower H_c component reflecting a recent viscous overprint. We interpret the high H_c magnetization as Late Cretaceous to Paleocene in age.

Neither the Maudlow Formation pole nor the pole derived from the high H_c overprint in the Livingston Formation agree with previously published poles for either the Late Cretaceous or Tertiary, but they do lie on a simple continuation of the APWP between Late Cretaceous and Paleocene mean poles. Secondary magnetizations found in the Franciscan Complex of California also yield poles which lie on a proposed extension of the APWP. If the Franciscan secondary poles represent an overprint event associated with early Tertiary accretion, and if the Livingston secondary pole represents a deformational event related to Laramide uplift of the nearby Beartooth Range, then together these three poles suggest that the North American APWP may continue farther to the east and south to define a Cretaceous-Tertiary cusp.

Introduction

The Late Cretaceous to early Tertiary part of the North American apparent polar wander path (APWP) is defined by only a few poles (Figure 1). The Paleocene and Eocene poles form groups which are not significantly different, but they are clearly distinct from the Late Cretaceous "stillstand" pole (Figure 1). The implication of this difference is a pronounced change in plate motion which may be correlated with important tectonic events in western North America. These include [1] the cessation of magmatism in the Sierra Nevada, [2] a decrease in the subduction angle of the Farallon Plate, [3] the onset of compressional deformation and the Laramide Orogeny, and [4] major changes in North American Plate motion with respect to the Farallon Plate and the hotspot reference frame. We present new results that may define the APWP in this important interval.

Geologic Setting

The Livingston and Maudlow Formations of southwest Montana (Figure 2) were chosen for study because their location to the east of the most disturbed part of the western Cordillera should produce undisturbed cratonic magnetizations, and the ages assigned to the formations are latest Cretaceous (Parsons, 1942; Marvin and Dobson, 1979; Langston, 1985; Skipp and McGrew, 1977; Skipp and Peterson, 1965; Vhay, 1939).

The Livingston Formation is located in the Crazy Mountains Basin of south-central Montana (45.5°N, 250.1°E). Our samples were taken from andesitic to dacitic welded tuffs or lapilli-tuffs in the lower part of the formation (Langston, 1985; Parsons, 1942; Vhay, 1939). The rocks are Campanian in age, based on stratigraphic correlation with sedimentary rocks in the basin, and a radiometric date of 79.8 ± 2.9 Ma (Tysdal *et al.*, 1986). The sampling area in the Livingston Formation is adjacent to the northern flank of the Beartooth Range, a Laramide foreland uplift block. The rocks closest to the front of the range were deformed into a series of folds, with deformation sharply decreasing with distance from the range front.

The Maudlow Formation (46.1°N, 248.9°E), is described in detail by Skipp and McGrew (1977) and Skipp and Peterson (1965). The formation consists of over 1500 m of intermediate-composition volcaniclastic rocks. In the area sampled, the Maudlow is surrounded by thrust faults, but these are all intraformational, which should produce little if any tectonic rotation, as discussed below. Many cooling units composed of welded tuffs, vitric and vitric-lithic tuffs, and flows are present throughout the section. The age of the Maudlow is latest Santonian to latest Campanian based on stratigraphic correlation with the Sedan Formation, and radiometric dates of 83 ± 2 Ma and 74.9 ± 1.0 Ma (Skipp and McGrew, 1977). This age assignment is also in agreement with the geomagnetic polarity changes discussed below.

Field and Laboratory Procedures

Fifty-one oriented samples from 11 sites were collected from the Livingston Formation and 66 oriented samples from 12 sites collected in the Maudlow Formation. The sites were located in folded strata to permit the application of a fold test. Structural attitudes of the sites were measured at the site, with the exception of

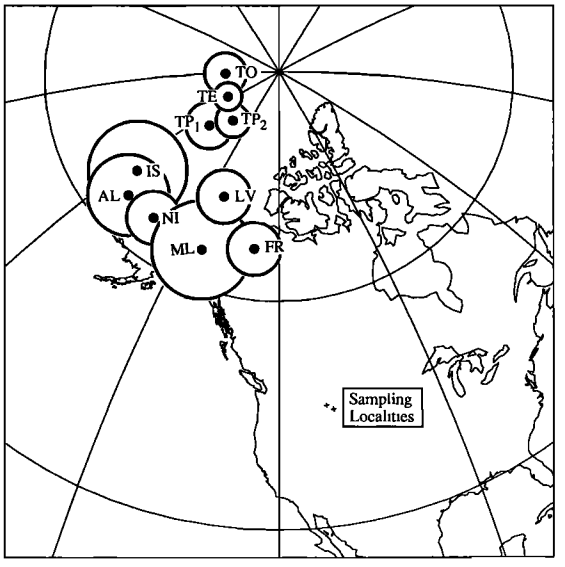


Fig. 1: Polar projection showing possible eastward extension of the Late Cretaceous North American APWP, error circles at P=0.33. Poles: [IS] Isachsen Diabase, Larochelle and Black (1963) and Larochelle et al. (1965); [AL] Arkansas alkalic rocks, Sharon and Hsu (1969); [NI] Niobrara Formation, Shive and Freirichs (1974); Maudlow [ML] and Livingston [LV] Formations (this paper); [FR] Franciscan Overprint, McWilliams (1988); [TP1] Paleocene average, Gordon (1984); [TP2] Paleocene (55-67 Ma), Diehl *et al.* (1983); [TE] Eocene (44-45 Ma), Diehl *et al.* (1983); [TO] Oligocene (22-38 Ma), Diehl *et al.* (1983).

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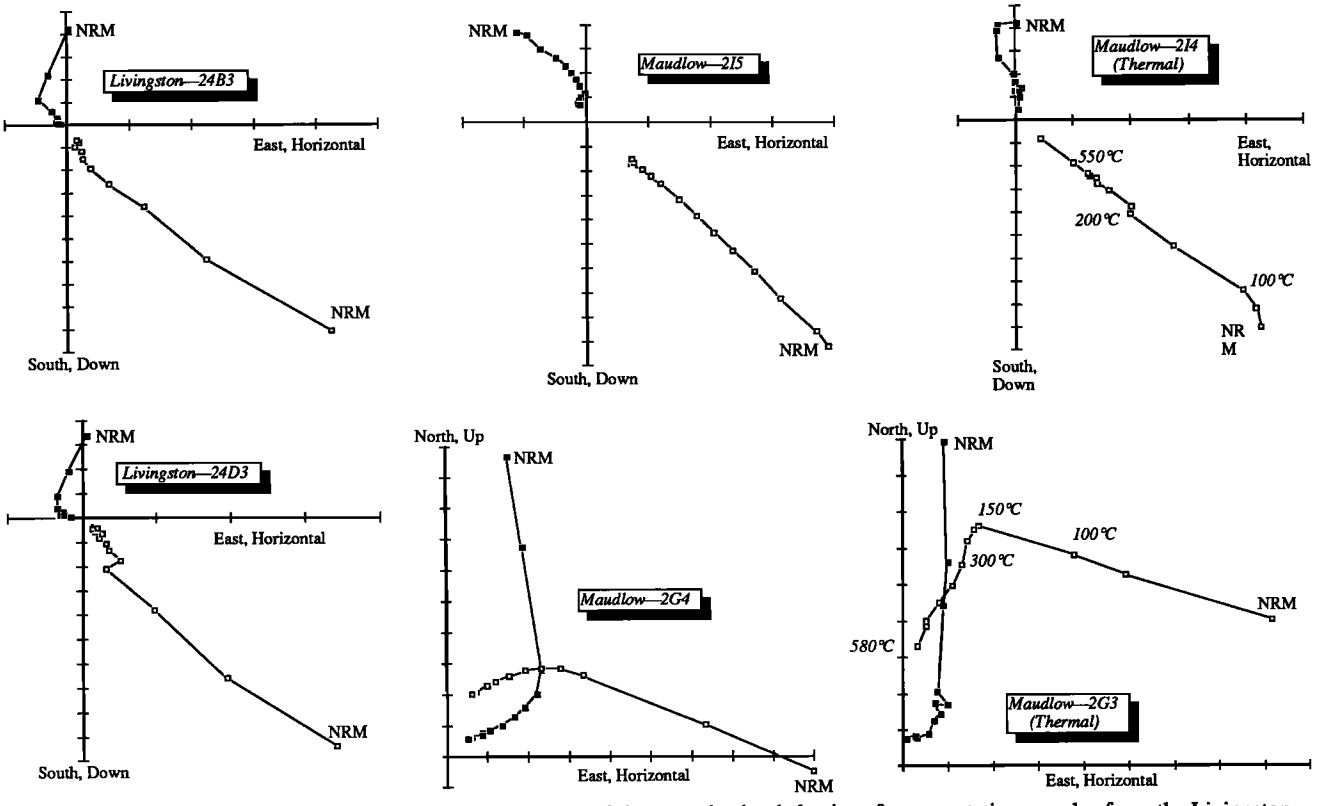


Fig. 2: Orthogonal vector diagrams showing AF and thermal demagnetization behavior of representative samples from the Livingston (AF only) and Maudlow Formations. Normal and reversed magnetizations are present in the Maudlow Formation. Two normal components are present in the Livingston Formation. Open (filled) symbols denote the *in situ* magnetization direction projected on the N-S vertical (horizontal) plane. Field increment is 5 mT in the upper four diagrams.

three sites in the Livingston Formation, where the attitudes were measured from a distance at which the general attitude could be seen. Overly weathered or altered rocks were avoided.

At least one specimen from each sample was subjected to progressive AF (alternating field) demagnetization with a peak field of 80 millitesla (mT), or progressive thermal demagnetization to peak temperatures of 700°C. AF demagnetization and thermal demagnetization were equally efficient in isolating a characteristic direction, thus the majority of the samples were cleaned using AF techniques. Characteristic directions were chosen using principal component analysis; sample mean directions were chosen from linear segments of the demagnetization path with an average of 5 points. Linear segments were not anchored to the origin.

Paleomagnetic Results

Livingston Formation

The Livingston rocks contain two magnetizations: a low H_c-low T_b component which is removed in fields less than 15 mT (and/or temperatures less than 100°C), and a high H_c-high T_b component that decays linearly to the origin (Figure 2; Table 1). The two

TABLE 1. Site Mean directions from the Livingston Formation

Site	N	D	I	a_{95}	k	DDir/Dip	D_c	I_c	Lat N	Lon E
24A	5	320.7	76.9	6.1	104	188/42	205.2	55.8	61.1	216.5
24B	3	348.4	74.3	3.3	601	339/38	342.2	36.5	73.4	230.0
24D	5	337.3	72.4	3.7	291	000/00	337.3	72.4	71.6	209.3
25A	4	333.3	77.9	9.3	57	201/43	216.4	54.2	64.5	225.9
25B	4	330.8	69.8	3.9	323	201/43	230.0	56.8	69.6	194.2
25C	5	340.9	74.8	6.1	106	201/44	218.8	56.5	70.6	222.1
25D	4	356.0	65.9	3.4	416	201/43	227.1	66.9	86.2	206.0
26C	4	351.5	70.3	4.7	218	030/20	9.8	52.6	79.6	221.7
26E	5	346.3	73.2	4.6	182	017/25	3.8	49.7	74.3	223.3
28A	4	342.1	75.7	4.1	289	348/18	345.3	57.7	69.8	226.3
Mean	10	341.9	73.4	2.6	289				72.3	218.3
Mean	10			18.4	6		291.3	75.4		

N: number of samples or sites; D, I, D_C, I_C: declination and inclination before and after structural correction; a_{95}: estimated semi-angle of the confidence circle at P=0.05; k: precision parameter estimate; DDir: Dip direction; Lat, Lon: coordinates of the virtual or geomagnetic pole (calculated from uncorrected directions).

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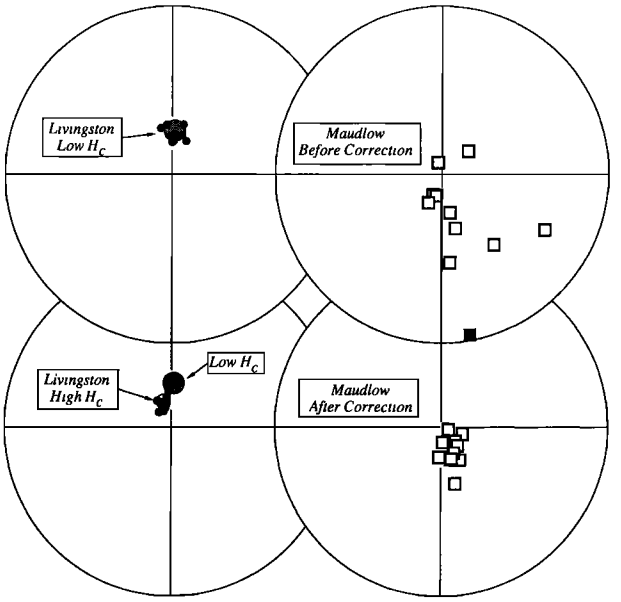


Fig. 3: Equal-angle stereographic projection of site mean directions. Left: low and high H_C force magnetizations from the Livingston Formation (shaded region is a_{95}). Right: decrease in scatter after tectonic correction, Maudlow Formation (normal samples inverted to upper hemisphere). Closed (open) symbols represent positive (negative) inclination. Mean directions omitted for clarity.

magnetizations are significantly different at the 95% confidence level. The low H_c (low T_b) component (I=64.1°, D=1.9°, N=10, a_{95}=4.5°) is indistinguishable from the dipole field direction (I=63.8°, D=0°) at the sampling location. High H_c directions at several sites are not significantly different from present day field or the dipole field; these were not used in calculating a paleomagnetic pole for the Livingston Formation.

The remaining sites display a high H_c—high T_b magnetization with a mean direction (I=73.4°, D=341.9°) which is too steep in inclination and too westerly in declination to be a present field (I=70.6°, D=13.3°), or dipole field overprint (Figure 3). This high H_c component fails the fold test, indicating a post-folding secondary origin (Table 1). A paleomagnetic pole calculated from virtual geomagnetic poles (VGPs) derived from these directions lies to the east and south of the Cretaceous, Paleocene and Eocene poles from elsewhere in North America (Figure 1).

Maudlow Formation

The Maudlow rocks also contain two magnetizations. A low H_c-low T_b component removed below 20 mT or 200°C is probably a recently acquired viscous magnetization. After treatment at higher temperatures, the samples display an essentially univectorial decay with both normal and reversed magnetizations (Figure 2). After inverting the directions of the reversed sites, the tectonically corrected mean direction (I=72.5°, D=334.6°) displays a nearly ten-fold improvement in the precision parameter, significant at the 95% confidence level (Figure 3; Table 2).

In making the tectonic correction, we rotated the Maudlow strata to paleohorizontal on the assumption that there was no initial dip. If an initial dip were present, this could produce an error in paleofield direction at a single site, but by combining results from different sites with differing tectonic attitudes we have attempted to reduce any systematic bias. Thus if initial dip errors are present in the Maudlow directions given in Table 2, they probably have increased the overall scatter without producing a large directional bias.

From stratigraphic control and K-Ar dating, the age of the Maudlow Formation spans latest Santonian to latest Campanian time (Skipp and McGrew, 1977). From these ages and the Kent and Gradstein (1985) time scale, we conclude that the reversal present in the section is Chron 33r. The positive fold test and good correlation with the polarity time scale both suggest that the magnetization is primary. VGPs from 11 sites were averaged and yield a paleomagnetic pole at 70°N, 208°E (A_{95}=9°). Like the Livingston pole, the Maudlow pole also lies to the east and south of the Cretaceous stillstand pole (Globerman, 1987) and other published Paleocene and Eocene poles (Figure 1).

Discussion

The poles calculated from the primary magnetization in the Maudlow Formation and the high H_c overprint in the Livingston Formation lie along a path that could be the continuation of the Cretaceous APW path (Figure 1). As discussed below, both lie near poles derived from secondary magnetizations of probable latest Cretaceous to Paleocene age from the Franciscan Complex of California (Figure 1; McWilliams, 1989).

The Maudlow pole is younger than all the poles that make up the Cretaceous stillstand average of Globerman (1987), yielding a suggestion that the Jurassic-Cretaceous APW track may have continued farther to the (present-day) east and south than suggested by May and Butler (1986), Gordon *et al.* (1984), or Diehl *et al.* (1980). An important factor in proposing this continuation is the key question of whether or not the Maudlow pole reflects an undisturbed paleofield direction. Many parts of southwest Montana have been affected by Laramide deformation, and it is possible that the Maudlow results have been rotated about a vertical axis. But Figure

TABLE 2. Site Mean directions from the Maudlow Formation

Site	N	D	I	a_{95}	k	DDir/Dip	D_c	I_c	Lat $_N$	Lon $_E$
30A	3	323.3	35.5	22.4	13	127/35	343.1	67.6	77.9	187.0
2F	6	339.8	-81.0	1.1	2757	357/28	184.9	-70.4	80.9	267.3
2G	4	48.3	-66.2	5.0	205	036/21	104.7	-84.8	47.7	234.3
2E	5	170.4	0.2	12.0	27	177/53	166.1	-52.3	73.1	113.0
2A	4	175.5	-34.5	9.7	52	186/35	160.6	-68.6	76.0	191.0
7A	5	117.7	-19.6	6.0	109	101/58	154.4	-70.6	71.6	196.9
2B	5	207.2	-74.8	3.1	415	246/08	175.4	-79.9	65.3	245.0
2C	5	195.3	-74.6	3.9	253	240/22	105.6	-74.8	46.2	207.5
2D	7	205.6	-68.7	3.5	234	240/22	134.5	-77.1	58.9	213.9
2H	6	345.8	54.4	8.6	44	200/14	329.8	64.8	69.3	172.2
2I	5	347.5	63.8	3.3	368	200/14	321.2	74.0	63.3	205.1
Mean	11	338.5	59.9	17.9	6					
Mean	11			5.7	55		334.6	72.5	69.8	207.8

N: number of samples or sites; D, I, D_C, I_C: declination and inclination before and after structural correction; a_{95}: estimated semi-angle of the confidence circle at P=0.05; k: precision parameter estimate; DDir: Dip direction; Lat, Lon: coordinates of the virtual or geomagnetic pole (calculated from the corrected directions). Sites listed in stratigraphic order.

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1 shows that the Maudlow pole is not rotated with respect to the other Late Cretaceous poles (IS, NI, AL); the main difference is in inclination, not declination. In the absence of significant error in estimating initial dip, tectonic rotation affects declination and not inclination. We conclude that the effects of tectonic rotation about a vertical axis are not discernable.

Deformation in the Beartooth Range, Crazy Mountains Basin and surrounding Laramide uplifts began as early as 78 Ma in the Madison Range of southwest Montana (Tysdal *et al.*, 1986), 75 Ma in the Tobacco Root Mountains (Schmidt and Garihan, 1983) and is contemperaneous with the end of the Sevier Orogeny in the Crazy Mountains Basin (Harlan *et al.*, 1988). The Beartooth Range had been uplifted and eroded to its Precambrian core by Paleocene time, indicating most deformation had occurred in latest Cretaceous times. Samples collected from the Livingston Formation were close to the uplifted front of the Beartooth Range, and thus during the Laramide orogeny, fluid circulation and high heat flow related to the these movements may have remagnetized the Livingston rocks, subject to the aforementioned timing constraints. A similar mechanism has been proposed for the wide spread syn-Alleghenian overprints found on the East coast (McCabe *et al.*, 1983, 1984, 1987; Oliver, 1986), in which fluid migration due to the orogeny overprinted lower Paleozoic rocks.

If this idea is correct, the high H_c overprint found in the Livingston rocks is of latest Cretaceous age and is related to the Laramide uplift of the neighboring Beartooth Range. The new Livingston pole is consistent with the hypothesis that the North American APW path moves southeastward following the 100-88 Ma Cretaceous stillstand (Figure 1), and lies just after a cusp in the APWP which may be related to a change in plate motion ultimately linked with the Laramide Orogeny. However, it must be remembered that the weight given to the Livingston pole is less than that given to the pole from the Maudlow Formation, since there is no firm paleohorizontal control for the Livingston overprint magnetization.

Secondary magnetizations with strongly negative fold tests from Franciscan and Great Valley Sequence rocks of California yield a paleomagnetic pole which also lies along the proposed southeastward extension of the APWP (McWilliams, 1989; Figure 1). These overprints are clearly not of recent origin. The Franciscan rocks which display the overprint have an inferred accretion time in Late Cretaceous to early Tertiary times (Tarduno *et al.*, 1985, 1986), whereas the Great Valley rocks were buried to a great depth and later uplifted in early Tertiary times (Frei, 1985). The overprints are probably Late Cretaceous to early Tertiary in age, and are probably linked to a thermal or chemical remagnetization event associated with the accretion process or uplift following protracted burial.

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References

Barnes, A.E. and R.F. Butler, A Paleocene paleomagnetic pole from the Gringo Gulch Volcanics, *Geophys. Res. Lett., 7,* 545—548, 1980.

Diehl, J.F. *et al.*, Paleomagnetism of the Late Cretaceous-early Tertiary north-central Montana alkalic province, *J. Geophys. Res., 88,* 10,593—10,609, 1983.

Diehl, J.F. *et al.,* Paleomagnetic results from early Eocene intrusions, north-central Montana: Implications for North American apparent polar wandering, *Geophys. Res. Lett., 7,* 541—544, 1980.

Frei , L.S., Paleomagnetic constraints on extension in the Basin and Range and in the North Atlantic area, Unpublished Ph.D Dissertation, Stanford University, 152—187, 1985.

Globerman, B.R., Investigation of two mid Cretaceous paleopoles from the North American craton: Arkansas and Yukon Territory, *Program and Abstracts, XIX General Assembly of the International Union of Geodesy and Geophysics, 2,* 501, 1987.

Gordon, R.G., A. Cox and S. O'Hare, Paleomagnetic Euler poles and the apparent polar wander and absolute motion of North America since the Carboniferous, *Tectonics, 3,* 499—537, 1984.

Harlan, S.S. *et al.*, Paleomagnetic and isotopic dating of thrust belt deformation along the eastern edge of the Helena Salient, northern Crazy Mountains Basin, Montana, Geol. Soc. America Bull., (in press).

Jacobsen, D. *et al.*, A Paleocene paleomagnetic pole for North America from alkalic intrusions, central Montana, *Geophys. Res. Lett., 7,* 549—555, 1980.

Kent, D.V. and F.M. Gradstein, A Cretaceous and Jurassic geochronology, *Geol. Soc. Am. Bull., 96,* 1419—1427, 1985.

Langston, R.B., Volcanic stratigraphy of part of the upper-Cretaceous Livingston formation near Nye, Montana, *Northwest Geology, 14,* 1985.

Marvin, R.F., and S.W. Dobson, Radiometric ages: Compilation B, United States Geological Survey, *Isochron/ West, 26,* 3—32, 1979.

May, S.R. and R.F Butler, North American Jurassic apparent wander: Implications for plate motion, paleogeography and Cordilleran tectonics, *J. Geophys. Res., 91,* 11,519—11,544, 1986.

McCabe, C. *et al.*, Diagenetic magnetite carries ancient yet secondary remanence in some Paleozoic sedimentary carbonates, *Geology, 11,* 221—223, 1983.

McCabe, C., R. Van der Voo, and M. Ballard, Late Paleozoic remagnetization of the Middle Ordovician Trenton Limestone, *Geophys. Res. Lett., 11,* 979—982, 1984.

McCabe, C., R. Sassen, and B. Saffer, Occurrence of secondary magnetite within biodegraded oil, *Geology, 15,* 7—10, 1987.

McWilliams, M., Late Cretaceous paleomagnetic overprint recorded in Franciscan Complex rocks, submitted to *Geology*, 1989.

Oliver, J., Fluids expelled tectonically from orogenic belts: Their role in hydrocarbon migration and other geologic phenomena, *Geology, 14,* 99—102, 1986.

Parsons, W.H., Origin and structure of the Livingston igneous rocks, Montana, *Geol. Soc. Am. Bull., 53,* 1175—1185, 1942.

Schmidt, C.J. and J.M. Garihan, Laramide tectonic development of the Rocky Mountain foreland of southwestern Montana, in: J. Lowell (ed.), *Rocky Mountains Foreland Basins and Uplifts,* Rocky Mountain Assoc. of Geol. Guidebook, 271—294, 1983.

Sheriff, S.D. and P.N. Shive, The Rattlesnake Hills of central Wyoming revisited: Further paleomagnetic results, *Geophys. Res. Let., 7,* 589—592, 1980.

Shive, P.N. and W.E. Freirichs, Paleomagnetism of the Niobrara Formation in Wyoming, Colorado, and Kansas, *J. Geophys. Res., 79,* 3001—3007, 1974.

Shive, P.N. and E.F. Pruess, A paleomagnetic study of basalt flows from the Absaroka Mountains, Wyoming, *J. Geophys. Res., 82,* 3039—3048, 1977.

Skipp, Betty, and L.W. McGrew, The Maudlow and Sedan Formations of the upper Cretaceous Livingston Group on the west edge of the Crazy Mountain Basin, Montana, *U.S. Geol. Survey. Bull., 1422-B,* 65p., 1977.

Skipp, Betty, and A.D.Peterson, Geological map of the Maudlow quadrangle, southwestern Montana, *U.S. Geol. Survey. Misc. Geol. Inv. Map I-452,* 1965.

Steiner, M.B. and C.E. Helsley, Reversal pattern and apparent polar wander for the late Jurassic, *Geol. Soc. Am. Bull., 36,* 1537—1543, 1975.

Tarduno, J., *et al.*, Franciscan complex Calera limestones-Accreted remnants of Farallon Plate oceanic plateaus, *Nature, 317,* 345—347, 1985.

Tarduno, J. *et al.*, Southern hemisphere origin of the Laytonville limestone of California, *Science, 231,* 1425—1428, 1986.

Tysdal, R.G., R.F. Marvin, and E. Dewitt, Late Cretaceous stratigraphy, deformation, and intrusion in the Madison Range of southwestern Montana, *Geo. Soc. Am. Bull., 97,* 859—868, 1986.

Vhay, J.S., Some features of the Livingston formation near Nye, Montana, *Trans. Am. Geophys. Union* , part 3, 433—437, 1939.

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